

A Novel Scene Matching Algorithm via Deep Learning for Vision-based UAV Absolute Localization

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Abstract—Almost all military systems and combat platforms rely on GNSS to achieve military missions. However, they are facing more and more intensive jamming and spoofing attacks. Due to the rapid development of computer vision technology and vision sensor, visual navigation and localization technology have gradually revealed their advantages to solve this kind of problem. In this paper, a new framework of visual localization under a GNSS-denied environment is proposed. Firstly, a novel image preprocessing method based on information entropy and granular segmentation is presented which can significantly filtrate the Google tiles in the georeferenced area to parsimony set to reduce UAV localization time. Secondly, we introduce an end-to-end image matching method based on convolutional neural network and image preprocessing to realize UAV localization which has strong robustness and localization accuracy. Finally, the proposed localization algorithm achieves experimental performance on the real images captured by DJI Phantom 3 drone. The good accuracy and fast execution time shown by experiments indicate that the proposed localization algorithm can serve as a powerful tool to keep the UAVs working continuously in the face of intensive jamming and spoofing attacks.

Keywords—UAV; deep learning; visual localization; image matching; image preprocessing

I. INTRODUCTION

Safe navigation is the key issue for UAVs to complete the combat mission and improve survivability. Nowadays, navigation for UAVs is usually using Global Navigation Satellite System (GNSS) [1], [2] or integrated navigation systems (INS) [3], typically composed of an INS and GNSS receiver. Their data are jointly processed in an optimal or suboptimal manner via some form of the Kalman Filter (KF) [3], [4]. Although modern INS/GNSS integrated systems can offer outstanding parameters and are adequate for numerous UAV applications, their inherent weakness consists of the lack of immunity of GNSS receivers to jamming. GNSS systems are vulnerable to interference from other radio signals, which results in lowered localization accuracy or even is unable to locate the position completely. The problem of jamming must be seriously considered especially for military systems.

In the case of signal loss, GNSS data are prolonged unavailable due to the jamming perpetually, which will constantly reduce the accuracy of the navigation system. Then the navigation for UAVs will only rely on INS. However, the accumulated errors without GNSS correction

derived from INS will be increased with time and distance traveled by the UAVs. And INS is not suitable for boarding on small UAVs because of its heavy weight.

In general, UAVs equipped in INS/GNSS systems do not possess the autonomous capacity required for most real military applications, and their missions can easily be interfered with by relatively simple GNSS jammers, commonly existed in contemporary battlefields. In recent years, the military community has been calling for multiple positioning, navigation, and timing (PNT) methods to complement and support GNSS, to guarantee UAVs working continuously in GNSS-denied environments (disturbed, spoofing, catastrophic failure).

To ensure the safe navigation task under a GNSS-denied environment, other navigation devices should be added to UAVs. Due to the rapid development of computer vision sensors and vision technology, visual-based technology has gradually shown its advantages to solve UAV navigation. Images can offer a wealth of information, and visual cameras are cheap and light enough to be mounted on board, so vision-based approaches can provide an excellent solution for UAV localization and navigation in a GNSS-denied environment. M. Jun proposed to use a lightweight visual-inertial system with a 2D georeference map to obtain accurate and consecutive geodetic positions for UAVs [20]. A. AL. Kaff presented a vision-based navigation (VBN) system based on the scene matching area correlation (SMAC), and experimental results demonstrated its availability for highly autonomous navigation of UAVs [5]. M. Shan proposed a UAV navigation system in a GPS-denied environment which used optical flow to predict its position in subsequent frames with HOG features for registration on Google Map [6]. Also, it utilized a particle filter to conduct a fine search for UAVs localization in which vision-based multiple usage localization algorithms were presented via the mutual information which was very robust for local and global scene variations, and the georeferenced images are used to solve the drift issue [7]. J. Kinnari proposed a GNSS-denied geolocalization of UAVs method by UAV image-to-map matching. This paper used the full image area of a UAV image instead of sparse feature points and it provided opportunity to utilize all available information in an image [18]. E. M. Tjahjadi proposed an image localization of single UAV images based on an orthogonal projection model. This paper used the measured ground control points and clustered the ground control points to perform feature-based UAV image registration. Then the UAV position was

estimated based on the results of image registration [8]. X. C. Liu proposed a non-GPS navigation problem by using visual sensors and georeferenced images [9]. X. L. and F. Q. Yu used the scene matching method to estimate the position and height of the UAV, and the flight test results showed that the methods could obtain the position information of the UAV accurately and efficiently [10], [11]. S. Sun proposed a twodimensional positioning method of aerial images based on scene matching [12]. Firstly, task planning was carried out to obtain aerial images of a fixed area. Then scene matching was completed based on a feature matching algorithm. The method of coordinate solution and base map generation was proposed to realize the two-dimensional localization of aerial photography. In [13], the Censur-star filter was firstly used to extract the feature points in the reference graph and realtime graph, and the FREAK binary descriptor was generated. Then, the Hamming distance was used as the similarity measurement of feature points, and the pairs of matching points were extracted by the distance ratio of k-nearest neighbor. Finally, the localization model based on RANSAC was used to obtain the spatial geometric transformation relation, to realize image matching, and obtain the longitude and latitude coordinates of locating point. A statistical approach was used to incorporate all available information. In [15], the fast image matching algorithm searched for an optimal scaling ratio to reduce the gap between resolutions of two images, and a validity index for validating the performance of matching was given. Finally, this algorithm obtained excellent matching results.

Therefore, most of the localization systems for UAVs based on visual information are the unsupervised ones, such as feature-based image matching. However, visionbased approaches depending on visual features are often very sensitive to illumination changes and environmental modifications.

Until now, only a very limited amount of deep learning networks has been done on UAV localization. Recently, M. Bianchi used a version-based method to localize a UAV by autoencoded Google map images-to-autoencoded UAV images matching. The autoencoder was a convolutional neural network, it was trained to compress images to a lowdimensional vector representation while retaining the key feature [?]. However, the architecture of the autoencoder has complex structure which needs a lot of training data support. Hunter and Simon proposed a localization method for UAVs to replace an onboard GPS system in the event of a noisy or unreliable GPS signal. The deep learning network used is a convolutional neural network based on VGG16. The input of the network is the UAV images and crops from the satellite

map. And the output of the network is warped and aligned parameters of the satellite map which will align the initial positions of UAVs. However, the accurate initial positions of UAVs and flight routes must be given in advance.

In this paper, to provide accurate position information of UAVs, deep learning technology with strong computing power is incorporated into the research of UAV visual localization to improve the real-time performance of visual localization. This absolute localization pipeline is based on 2-channel convolutional neural network [14]. The reasons for choosing the 2-channel network to establish the localization method network is relatively simple, which can make the image matching less time-consuming, and improve the real-time capability of UAV localization. Meanwhile, the UAV positioning method used in this paper can directly obtain the current UAV location information by matching the UAV images and the satellite tiles, without the complex optimization calculation and the collection of the UAV initial location, that will make the UAV localization high-efficiency. Furthermore, compared with the UAV positioning method mentioned above, the UAV positioning method proposed in this paper does not need to give the UAV flight route in advance. This method is expected to solve the problems of intensive interference and spoofing attack, and the autonomous safe navigation technology of UAVs will make great progress.

In summary, our contributions are as follows,

- An image preprocessing method based on image information entropy and granular segmentation is proposed which can increase the efficiency of UAV localization.
- A new visual localization scheme for UAVs in a GNSSdenied environment is proposed which first introduces 2-channel network to UAV visual localization. Compared with other visual localization methods, it achieves the higher accuracy and efficiency of UAV visual positioning in a GNSS-denied environment.
- This method can solve the matching failure caused by the large resolution difference between Google tiles and UAV images.

II. PROBLEM FORMULATION

The essence of visual localization is to load visual imaging modules on UAVs, and calculate the pose of UAVs via the camera imaging system and coordinate system. It takes UAVs as a target subject, and obtains real-time ground image in flight, then estimates UAV's positions and poses in spherical coordinate system through real-time image information.

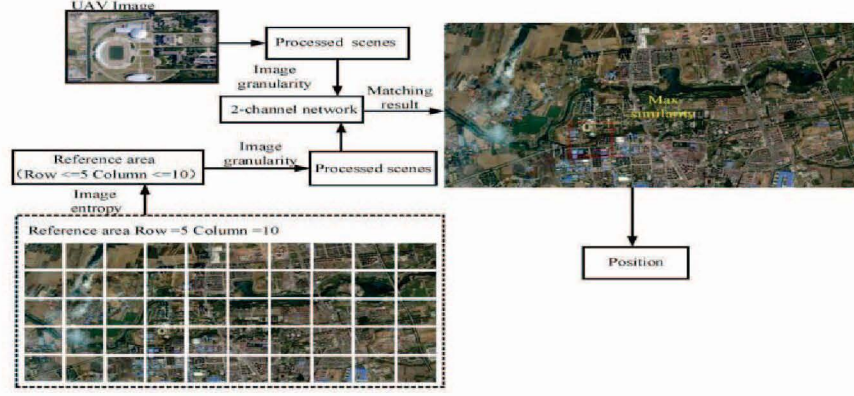


Figure 1. Vision-based localization system

The goal is to estimate the similarity S between a georeferenced image T_k and the real-time image of the UAV I_{UAV} . In the case of a similarity function f , the problem can be written as:

$$S = \underset{k}{\operatorname{argmax}} f(I_{UAV}, T_k) \quad (1)$$

where we search the S that maximizes the similarity between the satellites tiles $T_k, k = 1, 2, \dots, m$ in the georeferenced area and UAV image I_{UAV} .

To solve the maximization problem, the proposed visualbased localization pipeline for UAVs in a GNSS-denied environment is given in Figure 1. In order to improve the judgment performance of the similarity function, the satellites tiles T_k and I_{UAV} need to be preprocessed in advance. The image preprocessing consists of two parts which are image entropy and image granularity. And an image matching method based on the 2-channel convolutional network is proposed to judge the similarity between a georeferenced image T_k and UAV I_{UAV} . The detailed descriptions will be shown in the next section.

III. CONCLUSION

There are three key steps in the process of the proposed localization algorithm. The first one is to prune the tiles in the georeferenced area, so that the matched tiles will be reduced in order to enhance the localization efficiency. The second one is to established the granularity segmentation scheme to ensure integrity for matching. The third one is to image matching and position calculating. In Table 1, we list the key symbols used in the proposed localization algorithm.

A. Tiles Pruning

The pre-processing pipeline includes tiles pruning and image granularity. In order to reduce the computational burden, we will prune the satellite tiles which have low correlations with the matched UAV image. This procedure is rough filtering. Because the information entropy of one image represents a statistical form of features, which reflects the average amount of information in the image and represents the aggregation feature of distribution for image gray, this paper utilizes the information entropy to prune the satellite tiles.

TABLE I. KEY SYMBOLS USED IN THE PAPER

Symbol	Description
I_{UAV}	A shorthand of UAV image
T_k	A shorthand of the k-th satellites tiles in the georeferenced areas
E_{UAV}	The information entropies of UAV image
$E_n(k)$	The information entropies of the k-th satellites tiles
δ	A parameter to control the total numbers of tiles in Tselect
(i, j)	The label of the satellites tiles
U_v	A shorthand of the v-th sub-image of the segmented UAV image
$T_{u,i,j}$	A shorthand of the u-th sub-image of the segmented satellites tiles
$(U_v, T_v^{i,j})$	A matching pair as the network input
v, i, j, R_u	The result of the network
$m, R^{s,t}$	The maximum of the network output result
$T_u^{s,t,k}$	The corresponding tiles to the maximum network output result

Firstly, the information entropy EUAV of the UAV image IUAV is calculated, where

$$E_{UAV} = - \sum_{i=1}^{255} p_i \log p_i \quad (2)$$

and p_i represents the occurrence probability of the i -th gray value in the whole image matrix. Meanwhile, we calculate the information entropies $E_n(k), k = 1, 2, \dots, m$ of satellites tiles $T_k, k = 1, 2, \dots, m$ in the georeferenced area.

Then, if $E_{UAV} - \delta \leq E_n(k) \leq E_{UAV} + \delta$ the k -th tile T_k will be considered as a correlative element in T_{select} , to be matched with I_{UAV} where δ is a parameter to control the total numbers of tiles in T_{select} . The tiles pruning will reduce the computing burden for the localization system indirectly.

B. Image Granularity

The speed for matching is influenced by the resolutions of the matched images. Meanwhile, the closer the view of two images is, the higher the accuracy of image matching is. Therefore, in order to improve the matching performance, we propose a granular segmentation scheme to preprocess the UAV images and satellite tiles, in order to enhance the accuracies of image matching.

Firstly, the difference between the view of UAV image and Google tiles (e.g. the 15th level) can be roughly estimated according to prior knowledge, such as the level of satellite tiles, the resolution of the camera, and the flight height of UAVs. Then, the tiles and UAV images will be classified to different granularity octaves refer to Figure 2 and Figure 3. Finally, the segmented UAV images and Google tiles will be decreased the resolution to 64×64 to satisfy the input size of the network.

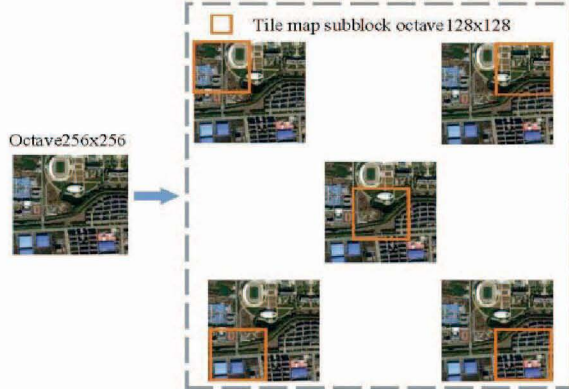


Figure 2. A granularity scheme for tile maps

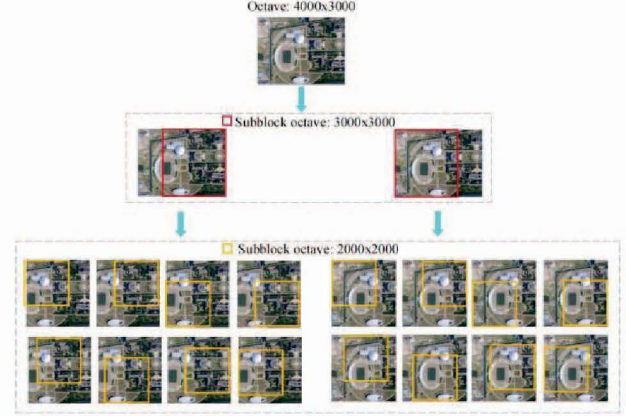


Figure 3. A granularity scheme for UAV images

In Figure 2, a tilemap with a resolution of 256×256 is divided into six 128×128 sub-blocks. The six sub-blocks have overlaps to ensure the accurate matching on the boundary of sub-blocks. Similarly, the granularity scheme for UAV images is proposed as shown in Figure 3. One image with resolution 4000×3000 captured by DJI Phantom 3 UAV is taken as an example to show the granularity scheme. The original image (the top one) is divided into two 3000×3000 sub-images. And then the sub-images will be disintegrated into 2000×2000 octaves. Finally, there will be 16 sub-blocks segmented from the original UAV image.

The matched pairs $(U_v, T_v^{i,j})$ are formed by the v -th subimage U_v segmented from the UAV image and u -th subimage of tile $T_v^{i,j}$, i and j are the label for the tile image $(U_v, T_v^{i,j})$ will be the inputs of the following network to obtain the location information.

C. 2-channel network architecture and learning

With the availability of large scale data sets and powerful computers for training, the vision community has recently seen a surge in the use of convolutional networks for various applications. It uses local receptive fields to extract local features of images and reduce the training parameters of the network, and it has the characteristics of weights sharing which can detect the same features in different positions of the image. What's more, it can remain the detected features translation invariant.

The CNN architecture consists of three primary parts: convolution layer, pooling layer and fully connected layer. The convolution layer is a linear, shift invariant operation that consists of performing local weighted combination across the input signal. The goal of the pooling operation is to control the amount of calculation by reducing network parameters and prevent the phenomenon of over fitting in the network. The fully connected layer is used to synthesize the extracted local features to obtain the global information of images.

2-channel network architecture is a fast accurate CNN [14], which is proposed by Simob. The input of 2-channel

network is a pair of the matching images. Because the 2-channel network has less training time and obvious advantages in network response time, compared with other deep learning networks. Therefore, a 2-channel network can be utilized to learn features, in order to solve the engineering applications for UAV localization.

The architecture of the 2-channel network is shown in Figure 4. The 2-channel network can be divided into three parts and it totally includes 7 layers.

- The first part is an input image layer. The input is considered to be a pair of image patches and the two patches of an input pair are defined as 2-channel images.
- The convolutional part includes three convolutional layers with two max-pooling layers. The kernels of the three convolutional layers are decreased gradually, which are 7, 5, 3. As the number of kernels is decreased, the detected features of the input images will be more clear. And each convolutional layer needs to be added a ReLU activation which can increase the nonlinearities. The max-pooling layer in the convolutional part aims to reduce data dimensions that can increase the speed of training the 2-channel network.
- The final part is the decision layer. The decision part consists of two fully-connected layers to do linear transformation, and the detected features and network outputs have a direct linear mapping through the fullyconnected layers.

In this paper, the model is trained in a supervised manner by using a hinge-base loss term and squared l_2 -norm regularization that leads to the following learning objective function.

$$\min_w \frac{\lambda}{2} \|w\|_2 + \sum_{i=1}^N \max(0, 1 - y_i o_i^{net}) \quad (3)$$

where w are the weight of the neural network, o_i^{net} is the net-

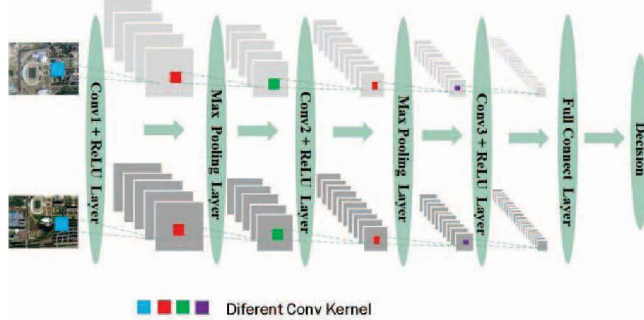


Figure 4. The structure of 2-channel network

work output for the i -th training sample, and $y_i \in \{-1, 1\}$ and the corresponding label with 1 and -1 respectively denoting a matching or a non-matching pair.

ASGD with constant learning rate 1.0, momentum 0.9 and weight decay $\lambda = 0.005$ is used to train the model. The mini-batches size of the model training is 128.

According to the Eq.(2), we can continuously optimize the network parameters and finally find a set of optimal network parameters, which can minimize the error between the output value of the network and the class value of the data.

This section first brings 2-channel network into UAVs localization which is conceptually suitable for solving the visual localization. Compared with other visual localization methods, it achieves the higher accuracy and efficiency of UAV visual positioning for GNSS-denied environment.

IV. EXPERIMENTAL RESULTS

A. Data Preparation

Experimental data preparation is a important component of the experiment. Our experimental data set consists of UAV image data and training image set. Different UAV images is indicated the different position of UAV. And the training image set is used to train the convolutional neural network. The specific instructions of two kinds data set are as followed.

1) *UAV images*: In this paper, the flight data were collected above the northern area of Shenyang. The original UAV images with resolution 4000×3000 are captured by a real drone (DJI Phantom 3). Flight altitude is approximately 500 meters high ($\pm 10m$). The pitch angle of the UAV camera is 0° . And the directions of the captured UAV images are consistent with Google maps. Figure 5 shows UAV images captured by DJI Phantom 3.

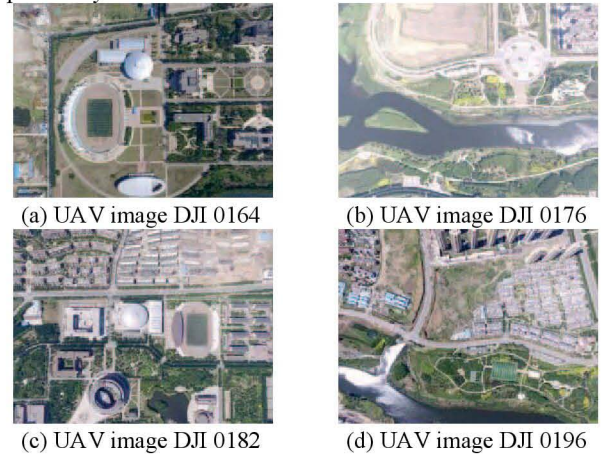


Figure 5. UAV images captured by DJI Phantom 3

We consider a reference template in Figure 6 of 1280×2560 pixels extracted from 15-level Google maps downloaded from Arc Tiler Downloader [16] in the region of the northern area of Shenyang, Liaoning, PR in summer. A tile map refers to a square grid image that segments a certain range of Google satellite maps into rows and columns according to a certain size, format, and scale. This georeferenced area includes 50 tiles (5 rows and 10 columns) as shown in Figure 7, each of which has a resolution 256×256 . The tiles in the same row or column have the same

latitude or longitude. The location information of each tile map can be obtained by the corresponding tile coordinate.

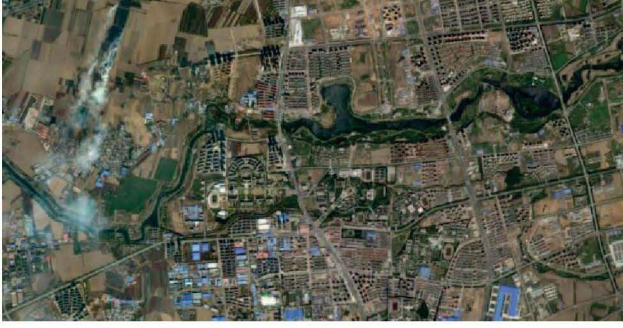


Figure 6. The Google map of the northern area of Shenyang

The proposed localization method in this paper is a solution for a UAV localization by matching the above sequence images taken from an on-board downward-looking camera with georeferenced tiles. However, the UAV images and georeferenced tiles should be preprocessed by the method in Section 2.1 to obtain the sub-images of UAV images and tiles.

2) *Training set*: The training dataset consists of Liberty and Notre Dame [17]. Each of the subsets contains more than 450000 image patches (64×64 pixels) sampled around the difference of Gaussian feature points, and the patches are scale and orientation normalized and includes the feature of rotational illumination scale changes. These training images were generated using actual 3D correspondences obtained via multi-view stereo depth maps. In order to increase the amount of training data set, the training data were augmented by flipping both patches horizontally and vertically and rotating to 90, 180, 270 degrees.

B. Results

The sub-image $T_u^{ij}, u = 1, \dots, 5$ and $U_v, v = 1, \dots, 16$ segmented from a UAV image U_v and a tile image T_u^{ij}



Figure 7. Tiles in the georeferenced area

in the georeferenced area are combined into a matching pair (U_v, T_u^{ij}) , which will be the input of the trained 2channel CNN. The output result vR_u^{ij} is a numerical value in $[-1, 1]$

which means the correct matching or incorrect matching between U_v, T_u^{ij} . The maximum output $mR_k^{s,t}$ will be determined by traversing all the tiles in georeferenced area, i.e., U_v is matched with $T_k^{s,t}$, if

$$mR_k^{s,t} = \max_{1 \leq i \leq 5, 1 \leq j \leq 10, 1 \leq u \leq 5, 1 \leq v \leq 16} \{vR_u^{i,j}\} \quad (4)$$

Then the location information of UAV can be obtained via the tile coordinate of Google tile $T_u^{s,t}$ which contains $T_u^{s,t}$ corresponding to the maximum value $mR_k^{s,t}$.

Finally, the tile coordinate of $T_u^{s,t}$ can transform to the location information with the longitude and latitude of UAV according to Eq.(4) and Eq.(5).

$$lon = \frac{tileX}{2^n} \times 360^\circ - 180^\circ \quad (5)$$

$$lat = \arctan(\sinh(\pi \times (1 - \frac{2 \times tileY}{2^n}))) \times \frac{180^\circ}{\pi} \quad (6)$$

where $tileX$ and $tileY$ are the tile coordinate and n is the level of the Google tiles (in this paper, $n = 15$).

We used the mainstream deep learning framework Pytorch to complete the coding of the whole training program. The deep learning network model was trained in a computer with a 2.90GHz Intel i7-10700f processor and 16GH RAM. And in order to improve the training efficiency, we used Nvidia 3070 GPU to accelerate the training process. Meanwhile, we used some OpenCV libraries to finish some simple image processing operations. Next, we will show the detailed localization results of flying DJI Phantom 3 UAV.

1) *UAV image DJI 0164*: The best tile matched with UAV image DJI 0164 is $T^{27614,12173}$ in Figure 8. And the current location information with longitude and latitude of UAV can be obtained according to Eq.(4) and Eq.(5), i.e., $(Lon, Lat) = (123.3764, 41.9268)$. Figure 9 shows the location of the current UAV (DJI 0164).

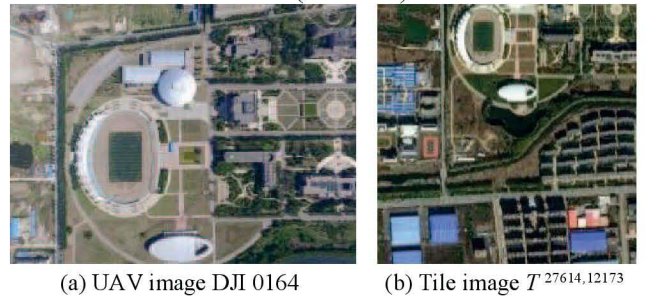


Figure 8. $T^{27614,12173}$ is the best matching with DJI 0164

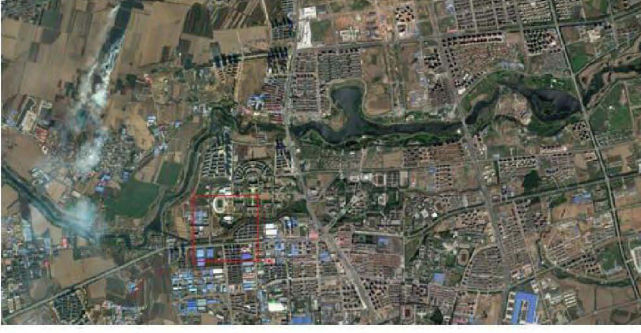


Figure 9. The red box represents the location of the current UAV in the georeferenced area.

2) *UAV image DJI 0176*: The best tile matched with UAV image DJI _0176 is $T^{27617,12171}$ in Figure 10. And the current location information with longitude and latitude of UAV can be obtained according to Eq.(4) and Eq.(5), i.e., $(Lon, Lat) = (123.4094, 41.9431)$.

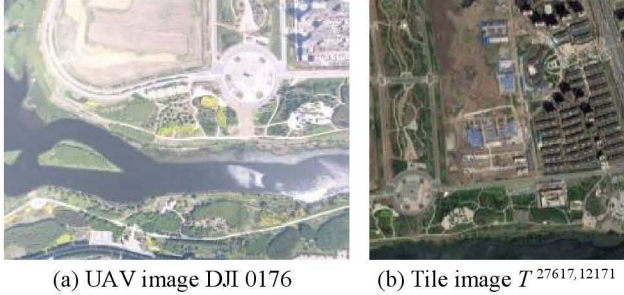


Figure 10. $T^{27617,12171}$ is the best matching with DJI 0176

3) *UAV image DJI 0182*: The best tile matched with UAV image DJI _0182 is $T^{27616,12172}$ in Figure11. And the current location information with longitude and latitude of UAV can be obtained according to Eq.(4) and Eq.(5), i.e., $(Lon, Lat) = (123.3984, 41.9349)$.

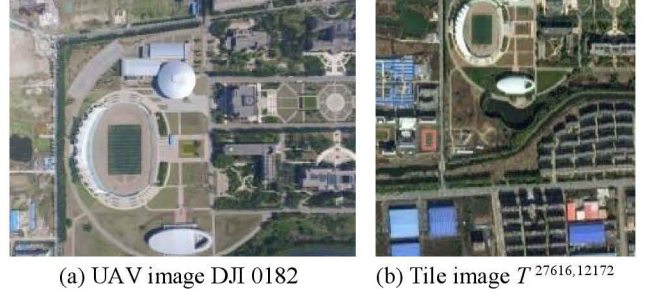


Figure 11. $T^{27616,12172}$ is the best matching with DJI 0182

4) *UAV image DJI 0196*: The best tile matched with UAV image DJI 0196 is $T^{27619,12171}$ in Figure 12. And the current location information with longitude and latitude of UAV can be obtained according to Eq.(4) and Eq.(5), i.e., $(Lon, Lat) = (123.4313, 41.9431)$.



Figure 12. $T^{27619,12171}$ is the best matching with DJI 0196

C. Localization Performance

In this section, we collect the localization precision of the proposed algorithm in Table 2. It lists the longitude and latitude of GPS position, the corresponding positions obtained via our algorithm, and the errors on longitude and latitude.

TABLE II. LOCALIZATION PRECISION

UAV Image	GPS position (°)	Localization position(°)	Error (°)
(a) DJI 0164	(123.3763, 41.9236)	(123.3765, 41.9268)	(-0.0002, -0.0032)
(b) DJI 0176	(123.4030, 41.9330)	(123.3984, 41.9431)	(0.0046, -0.0101)
(c) DJI 0182	(123.3998, 41.9262)	(123.3984, 41.9350)	(0.0014, -0.0088)
(d) DJI 0196	(123.4286, 41.9395)	(123.4314, 41.9431)	(-0.0028, -0.0036)

Furthermore, the proposed method is compared with the feature-based image matching method and 2-channel network. SURF algorithm is a traditional feature points extraction algorithm. It is one of the most widely used image matching algorithm [15]. Meanwhile, in order to show the efficacy of the image preprocessing in the proposed algorithm, we compare the experimental results with 2channel network. To highlight the advantage of the proposed absolute localization techniques, we present the accuracy and the execution time of three algorithms in Table 3. In Table 3, the average time for matching an image pair,

and the accuracy means the proportion of correct matching for UAV images. From Table 3, it can be seen that the proposed method obtained the highest accuracy with the least matching time. And the preprocessing for UAV images and tilemaps plays a key role in improving the accuracy and time consumption of localization.

TABLE III. COMPARISONS ON TIME AND ACCURACY

Algorithm	Time	Accuracy(%)
SURF based image matching	0.24	50.0
2-channel	0.029	50.0
The proposed method	0.023	100.0

Meanwhile, we also investigate the efficacy of the preprocessing method proposed in this paper. We remove granularity segmentation and observe the accuracy of the same testing images. The accuracy drops from 100% to 50%. And the influence of information entropy in our method is also analyzed. Experiments on the testing UAV images show an increase of execution time from 0.023s to 0.029s with nearly 26.1% of the proposed method. This result proves the advantage of the preprocessing method.

Comparing to existing work, the proposed method requires no prior knowledge of the initial position and achieves precise localization without using any auxiliary equipment. Therefore, the work will be expected to solve intensive interference and spoofing attack during the UAVs autonomous safe navigation.

V. CONCLUSION

This paper presented a promising image matching framework based on deep learning for UAV localization. The main contributions fall into two parts. First, using a convolutional neural network to realize the matching of high-resolution UAV images and low-resolution satellite images. And an image preprocessing based on image entropy and granularity segmentation methods is proposed to finish the selection and formation of image matching pairs. The image entropy can significantly decrease the UAV localization time by eliminating satellite tiles with a larger information entropy gap with the UAV images. And the image granularity was presented to solve the resolution gap problem between UAV images and satellite tiles which can improve UAV localization accuracy. Comparisons on the experimental results with other localization methods, our method has significant advantages in localization precision and localization time. More efforts will be devoted to the promotion of the UAV localization method in four seasons (especially in winter) and night.

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